

Library: A Large-Scale Outdoor Gaussian Splat Reconstruction Dataset

Reagan Koo*
SungkyunKwan University

Yeong-Gyu Kim†
SungkyunKwan University

Isaac Yang‡
SungkyunKwan University

Seung Huk Ahn§
SungkyunKwan University

Eun-Seok Ryu¶
SungkyunKwan University



Figure 1: Large-scale outdoor reconstruction of a university library and its surrounding campus environment learned from the Library dataset. The scene is reconstructed using Gaussian Splatting from drone-captured videos, demonstrating coherent building-scale geometry and appearance across a wide outdoor area.

ABSTRACT

Large-scale outdoor scenes captured by drones exhibit wide spatial extent and diverse viewpoint distributions, making training view density an important factor in view synthesis. While recent Gaussian Splatting-based methods have demonstrated efficient real-time rendering, their behavior under varying numbers of training views in real-world, building-scale scenes remains insufficiently explored.

This work introduces a large-scale drone-captured dataset of a university library and its surrounding area, which has been adopted as a representative large-scale outdoor sequence in the MPEG GSC common test conditions (CTC) to support standardized evaluation of Gaussian Splatting-based reconstruction in outdoor environments. The dataset was captured along circular flight trajectories at multiple altitudes and distances, providing systematic coverage of near-range and far-range viewpoints. Using a fixed set of evaluation views, the results show that faithful large-scale scene coverage can be achieved with as few as approximately 200 training images, despite the wide spatial extent of the outdoor environment.

Index Terms: Large-Scale Scene Reconstruction, Gaussian Splatting, Drone-Captured Dataset, View Synthesis.

*e-mail: reagan9@skku.edu

†e-mail: kyk2798@skku.edu

‡e-mail: icarax1491@skku.edu

§e-mail: powerminers@skku.edu

¶e-mail: esryu@skku.edu

1 INTRODUCTION

Recent learning-based view synthesis methods have enabled high-quality novel view rendering through increasingly efficient scene representations. Among them, Gaussian Splatting [4] has emerged as a practical approach that combines competitive visual quality with real-time rendering performance, making it suitable for large-scale scene reconstruction. When applied to large-scale outdoor scenes captured by drones, view synthesis operates under conditions that differ from small-scale or object-centric settings. Drone-based acquisition typically produces long video sequences covering wide spatial areas, with viewpoints distributed along continuous flight trajectories. In such scenarios, the number and distribution of training views become important factors in determining reconstruction behavior, while overly dense sampling may introduce unnecessary computational and memory overhead.

The effect of training view density in large-scale outdoor environments has not been sufficiently analyzed using datasets that reflect realistic drone capture conditions. This tendency has also been observed in MPEG, where standardization efforts for Gaussian-based scene representations have primarily relied on indoor or limited-scale multiview datasets such as Cinema[2], Breakfast[9], VRrooms[1], and Bartender[3] for evaluation. As a result, it has remained difficult to systematically analyze how variations in training view count affect reconstruction behavior in building-scale outdoor scenes. This paper introduces the Library [5] dataset, which has been adopted as a large-scale outdoor sequence in the MPEG GSC common test conditions (CTC). The dataset covers a building-scale outdoor environment captured along real drone flight trajectories and includes a fixed set of evaluation views, enabling controlled analysis of Gaussian Splat-based reconstruction behavior under varying numbers of training views in outdoor settings.



Figure 2: Drone-based data acquisition around the library building. The left image shows a drone capturing the building façade while flying along a controlled circular trajectory at a fixed altitude. The right image shows the live camera view displayed on the drone controller during flight, illustrating the aerial capture of the surrounding outdoor environment with consistent viewpoint coverage.

2 RELATED WORK

2.1 3D Gaussian Splatting

Gaussian Splatting [4] represents scenes using explicit 3D Gaussian primitives optimized through differentiable rendering. By directly modeling spatial primitives and their radiance properties, this approach enables efficient optimization and real-time rendering while maintaining competitive visual quality. Due to these properties, Gaussian Splatting has been adopted in a variety of interactive and large-scale rendering scenarios. In practice, gsplat [11], a publicly available open-source implementation, provides an optimized training and rendering pipeline that facilitates scalable Gaussian Splat-based reconstruction on modern GPU architectures.

2.2 Large-Scale Scene Reconstruction

Following earlier studies on large-scale neural scene reconstruction based on NeRF representations [8, 10, 12], recent work has begun to investigate the scalability of 3D Gaussian Splatting for outdoor and urban environments. CityGaussian [7] explores large-scale reconstruction through a divide-and-conquer training strategy combined with level-of-detail representations, while VastGaussian [6] addresses scene-scale complexity via progressive spatial partitioning and parallel optimization. These studies indicate that Gaussian-based explicit representations can be extended beyond object-centric settings to support efficient reconstruction and real-time rendering in building- and city-scale outdoor scenes.

3 METHODS

3.1 Data Acquisition

The dataset is captured on a university campus under controlled conditions, focusing on the main library building and its surrounding outdoor environment. Figure 2 shows that data acquisition is performed using a DJI Mavic 3 Pro drone equipped with a telephoto camera module. The camera features a 1/2-inch CMOS sensor with 12 megapixels and a 166 mm focal length lens (35 mm equivalent) with an $f/3.4$ aperture, providing a 15° field of view. All videos are recorded at 4K resolution using this fixed camera configuration, ensuring consistent intrinsic parameters throughout the capture process.

As illustrated in Figure 3 the drone follows controlled circular flight trajectories around the target building to provide systematic viewpoint coverage. Data is captured at two flight altitudes, 50 m and 70 m, producing both near-range and far-range observations of the structure. The total capture area covers approximately 98,580 m^2 , with a perimeter of about 1,030 m. To increase viewpoint diversity and redundancy, the same capture configuration is repeated across four independent flight sessions, resulting in a large collection of video frames sampled from continuous trajectories. Minor dynamic elements such as moving vehicles are occasionally present

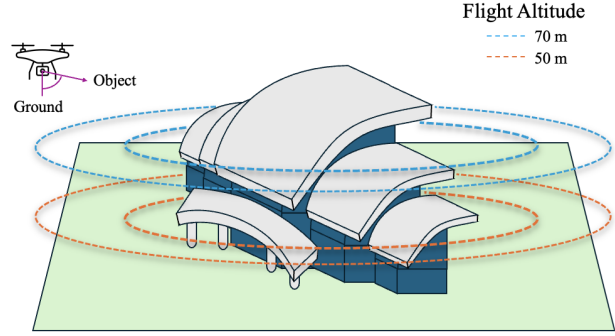


Figure 3: Illustration of the drone flight paths used for data acquisition around the library. The drone follows circular trajectories at two altitudes, 70 m and 50 m above the ground, corresponding to farther and nearer viewpoints, respectively, while the camera continuously faces the building. This setup captures both near-view and far-view observations of the target structure.

but occupy a negligible portion of the scene and do not affect reconstruction quality. All sequences are captured under stable daylight conditions, and no additional color correction is applied during pre-processing.

Video sequences are subsequently converted into image frames using uniform temporal sampling. Extracted frames are resized to a fixed spatial resolution and organized according to capture altitude and viewing distance, forming a common frame pool that preserves the original camera motion and viewpoint distribution. This frame pool serves as the basis for constructing controlled training and evaluation subsets used in the experiments.

3.2 Dataset Construction and Splitting

3.2.1 View Sampling Strategy

From the drone-captured videos, a total of 173,292 image frames are extracted, forming a large common frame pool. All videos are originally recorded at 4K resolution, but for all experiments, extracted frames are downsampled to Full HD (1920×1080) to reduce computational cost. The images are grouped according to predefined capture configurations based on flight altitude (50 m and 70 m) and viewing distance (near and far). Each group contains several tens of thousands of frames sampled along continuous circular flight trajectories, providing dense and redundant viewpoint coverage.

Training sets are then derived from this common pool by progressively reducing the number of input views. Specifically, training view counts of 50, 100, 200, 400, 800, and 1,600 images are considered. All training sets are constructed as nested subsets, such that each larger set strictly includes all images from the smaller ones. To ensure consistent viewpoint distribution across experiments, the total number of training views is evenly allocated across altitude and distance groups (50 m / 70 m, near / far), and frames within each group are selected using uniform sampling along the original temporal order of the videos.

3.2.2 Test Set Construction

A fixed test set of 200 images is selected prior to training and shared across all experiments. Test images are uniformly sampled from each capture group and explicitly excluded from the training frame pool to prevent any overlap between training and evaluation data. For distance-dependent evaluation, the test set is divided into near-view and far-view subsets. Near-view test images correspond to viewpoints captured at closer distances to the building facade, while far-view test images represent viewpoints acquired

Table 1: Reconstruction performance under different training view counts on the Library dataset. Results are reported separately for far-view, near-view, and their combined (total) test sets. PSNR is reported as RGB PSNR in decibels (dB), LPIPS is computed using the AlexNet backbone, and model size is reported in megabytes (MB).

Views	Far			Near			Total			Size
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	
50	22.08	0.6907	0.2319	25.54	0.7960	0.1429	23.82	0.7438	0.1876	599
100	25.15	0.7993	0.1737	28.20	0.8742	0.1115	26.70	0.8372	0.1429	611
200	28.74	0.8737	0.1452	30.00	0.9084	0.1026	29.41	0.8915	0.1243	597
400	28.47	0.8705	0.1468	29.90	0.9050	0.1061	29.22	0.8882	0.1268	605
800	28.72	0.8749	0.1467	29.94	0.9063	0.1062	29.36	0.8911	0.1269	594
1600	28.74	0.8757	0.1451	30.01	0.9065	0.1057	29.39	0.8916	0.1258	599

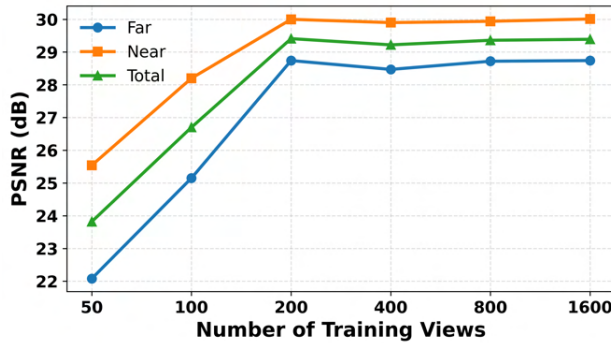


Figure 4: Reconstruction performance as a function of the number of training views on the Library dataset. The plot shows PSNR measured on the fixed test set as training view count increases from sparse to dense regimes, illustrating rapid performance improvement at low view counts followed by early saturation.

from larger stand-off distances that capture the overall building context. This separation enables a direct comparison of reconstruction behavior under different viewing distances while keeping the evaluation views fixed across all training configurations.

3.2.3 Training Setup

All models are trained using *gsplat*, a GPU-efficient implementation of Gaussian Splatting designed to support large-scale scene reconstruction. Given the building-scale extent of the Library dataset and the large number of Gaussian primitives involved, *gsplat* enables efficient memory usage and stable optimization during training.

Compared to the default training configuration, the optimization schedule is extended to better accommodate the complexity of the large outdoor scene. Specifically, the densification phase is prolonged from 15,000 to 30,000 iterations, and the total number of training iterations is increased from 30,000 to 60,000 iterations. This adjustment allows the model to more fully capture fine-scale geometric and appearance details while maintaining stable convergence in large-scale settings.

For evaluation, validation is performed exclusively using the fixed test set selected prior to training, ensuring strict separation between training and evaluation data. No training images are reused during validation. All quantitative results reported in this paper are computed based on renderings from this predefined test set, allowing fair comparison across different training view counts.

4 RESULT

This section presents the reconstruction results obtained under different training view counts on the Library dataset. The goal is to examine how reconstruction quality evolves as the number of input views increases in a large-scale outdoor setting with substantial viewpoint and appearance variation.

Table 1 and Figure 4 summarize the quantitative reconstruction performance as a function of training view count. Both the tabulated results show that reconstruction quality increases rapidly as the number of training views grows from 50 to 200, after which the improvement rate significantly slows down. For the combined test set, PSNR reaches 29.41 dB with only 200 training views, while further increasing the training set size up to 1,600 yields only marginal gains. Figure 5 presents a qualitative comparison of rendering results for models trained with 200 and 1,600 views.

This trend is consistently observed for both near-view and far-view evaluations. While near-view reconstruction benefits slightly more from increased training density, far-view performance follows a similar saturation pattern, suggesting that the dominant global structure and appearance of the scene are already captured with a limited number of well-distributed views. These results highlight that the proposed dataset enables effective large-scale reconstruction without relying on excessively dense view sampling.

5 CONCLUSION

This paper introduces the Library dataset, a large-scale outdoor dataset designed for analyzing Gaussian Splat-based reconstruction under realistic drone-captured settings. The dataset encompasses a wide range of challenging visual phenomena commonly encountered in real-world outdoor environments, including non-Lambertian reflections from glass façades, strong viewpoint-dependent appearance variations induced by multi-altitude capture, and high-frequency geometric and radiometric details arising from fine-scale structures such as trees and vegetation. By jointly including near-view and far-view observations, the dataset captures both local reconstruction fidelity and global structural consistency within a single scene. Experimental results further demonstrate that the essential geometric structure and appearance characteristics of a large-scale environment can be effectively captured using a moderate number of well-distributed training views. These properties position the Library dataset as a practical and versatile resource for studying Gaussian Splat-based reconstruction in complex, large-scale outdoor environments.

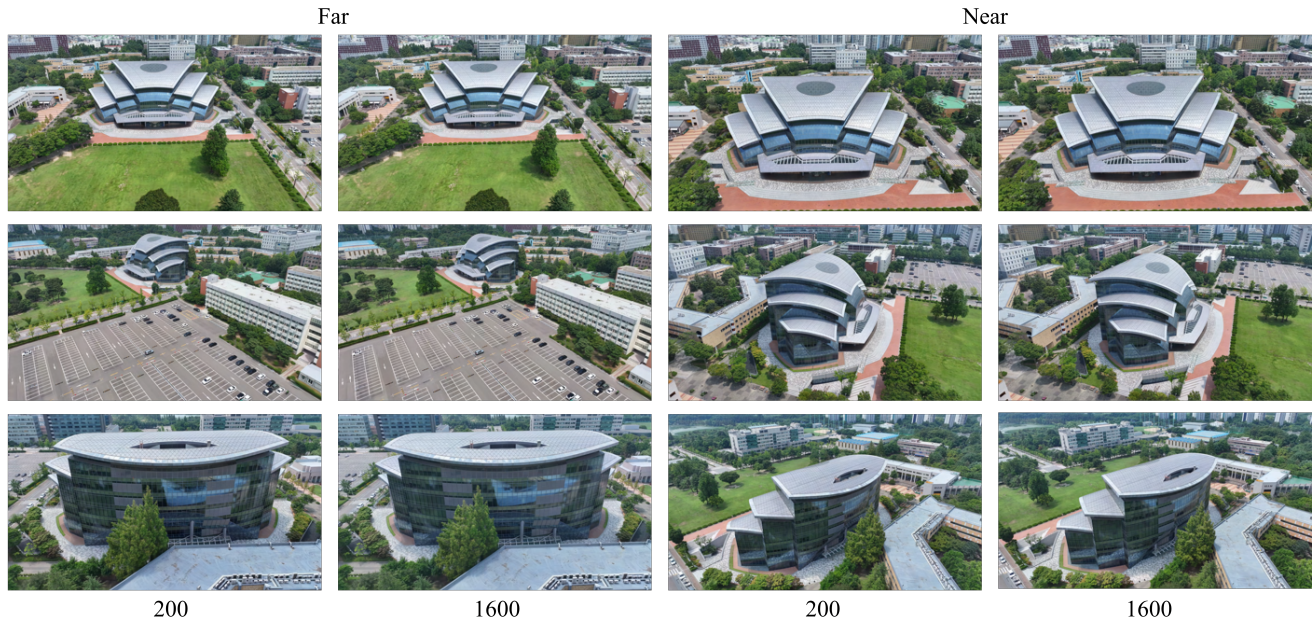


Figure 5: Qualitative comparison of reconstruction results on the Library dataset under different training view counts. Results are shown for far-view (left) and near-view (right) test viewpoints. For each setting, reconstructions trained with 200 views (left) and 1,600 views (right) are compared using the same fixed test views. While increasing the number of training views improves fine-scale details, the overall scene structure and appearance are already well captured with 200 training views.

6 ACKNOWLEDGMENT

This work was supported by Electronics and Telecommunications Research Institute (ETRI) grant funded by ICT R&D program of MSIT/IITP[2022-0-00022-002, Development of immersive video spatial computing technology for ultra-realistic metaverse services]

This research was also supported by the "Regional Innovation System & Education (RISE)" through the Seoul RISE Center, funded by the Ministry of Education (MOE) and the Seoul Metropolitan Government. (2025-RISE-01-018-05)

REFERENCES

- [1] J. Choi, Y. Ryu, Y. Choi, J.-B. Jeong, J.-H. Park, I. Yang, and E.-S. Ryu. Vrooms: Multiview datasets for radiance fields experiments. In *Proceedings of the 37th Workshop on Image Processing and Image Understanding (IPIU)*, 2025. 1
- [2] J. Y. Jeong, J. Kim, B. H. Lee, K. J. Yun, W. Cheong, and S. H. Yoo. [INVR] Multiview dataset Cinema for 3D INVR activity. Technical Report m69932, ISO/IEC JTC 1/SC 29/WG 4, Online, January 2025. 1
- [3] J. Y. Jeong, R. Koo, K. Oh, H. Shin, and G. Lee. [GSC][JEE 6.1-related] Training Method for Generating Temporally Consistent Per-frame I-3DGS Dataset. Technical Report m71763, ISO/IEC JTC 1/SC 29/WG 4, Online, January 2026. 1
- [4] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), July 2023. 1, 2
- [5] R. Koo, Y.-G. Kim, I. Yang, S. H. Ahn, and E.-S. Ryu. Proposals of library sequence. Technical Report MPEG2025/m74011, ISO/IEC JTC1/SC29/WG4 MPEG, 2025. 1
- [6] J. Lin, Z. Li, X. Tang, J. Liu, S. Liu, J. Liu, Y. Lu, X. Wu, S. Xu, Y. Yan, and W. Yang. Vastgaussian: Vast 3d gaussians for large scene reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5166–5175, June 2024. 2
- [7] Y. Liu, C. Luo, L. Fan, N. Wang, J. Peng, and Z. Zhang. Citygaussian: Real-time high-quality large-scale scene rendering with gaussians. In *European Conference on Computer Vision*, pp. 265–282. Springer, 2024. 2
- [8] M. Tancik, V. Casser, X. Yan, S. Pradhan, B. Mildenhall, P. P. Srinivasan, J. T. Barron, and H. Kretschmar. Block-nerf: Scalable large scene neural view synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8248–8258, 2022. 2
- [9] T. Tapie, A. Schubert, R. Gendrot, G. Briand, F. Thudor, and R. Doré. Breakfast new natural content proposal for MIV. Technical Report m56730, ISO/IEC JTC 1/SC 29/WG 4, Online, April 2021. 1
- [10] H. Turki, D. Ramanan, and M. Satyanarayanan. Mega-nerf: Scalable construction of large-scale nerfs for virtual fly-throughs. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12922–12931, 2022. 2
- [11] V. Ye, R. Li, J. Kerr, M. Turkulainen, B. Yi, Z. Pan, O. Seiskari, J. Ye, J. Hu, M. Tancik, et al. gsplat: An open-source library for gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025. 2
- [12] X. Zhang, S. Bi, K. Sunkavalli, H. Su, and Z. Xu. Nerfusion: Fusing radiance fields for large-scale scene reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5449–5458, 2022. 2